Patuakhali Science and Technology University

Department of computer Science and Information Technology

6th Semester (Level-3, Semester-II), **Final Examination** of B.Sc. Engg. CSE, July-December/2022

Session: 2019-20

Course Code: CIT-321 Course Title: Operating System

Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks]

Answer any 5 of the following questions.

1\. (a) Define cooperating process. Develop a model of a system consisting of cooperating sequential processes or threads with the producer-consumer problem. 4

Chapter -6

(b) i) What is critical-section problem? Explain the requirements which must be satisfied to solve the critical-section problem. Review the limitations of semaphore solution to the critical section problem. 7

ii) Explain the strategy, high-level synchronization construct-the monitor type, for dealing with such errors.

(c) What is spinlock? State the effect of liveness failure in the process execution life cycle. 3

2\. (a) Write a formal definition of deadlock in a multithreaded application. Point out the necessary conditions, which hold simultaneously in a system, for deadlock situation. 4

Ch-7

(b) Illustrate and explain the resource-allocation graph, resource-allocation graph with deadlock, and resource-allocation graph with a cycle but no deadlock. 5

(c) How to detect deadlock in a system? Illustrate and explain the data structures, which must be maintained, process, and example of the deadlock detection algorithm with several instances of each resource type. 5

3\. (a) Formulate the readers-writers problem as a classical synchronization problem. Discuss the solution of the problem using binary and mutex semaphore. 4

(b) Define and demonstrate the dining-philosophers problem as a synchronization problem. State the semaphore and monitor solutions of the dining-philosophers problem. 4

Ch-8

(c) Describe a mechanism for enforcing memory protection in order to prevent a program from modifying the memory associated with other programs. 3

(d) Compare and contrast monolithic, layered, microkernel, modular, and hybrid strategies for designing operating systems. 3

4\. (a) Define operating system. Identify services provided by an operating system. 3

Ch-1 Ch-2

(b) What is system call? Illustrate how system calls are used to provide operating system services. - Ch-2 4

Ch-2

(c) Why applications are operating system specific? Identify the separate components of a process and illustrate how they are represented and scheduled in an operating system. 3

-3

(d) -3 Define process. Describe how processes are created and terminated in an operating system, including developing programs using the appropriate system calls that perform these operations. 4

5\. (a) 3 What is process control block (PCB)? Describe and contrast inter-process communication using shared memory and message passing. 3

(b) -4 Define thread. Identify the basic components of a thread, and contrast threads and processes. 4

(c) 4 Draw multi-threading models. Describe the major benefits and significant challenges of designing multi-threaded processes. 3

(d) 4 Describe about Java Threads. Illustrate different approaches to implicit threading, including thread pools, fork-join, and Grand Central Dispatch. 4

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6\. 5 (a) Why CPU scheduling needed? Describe various CPU scheduling algorithms. 2

(b) Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. In answering the questions, use nonpreemptive scheduling, and base all decisions on the information you have at the time the decision must be made. 5

5

Process	Arrival Time	Burst Time
P1	0.0	8
P2	0.4	4
P3	1.0	1

i) What is the average turnaround time for these processes with the FCFS scheduling algorithm?
ii) What is the average turnaround time for these processes with the SJF scheduling algorithm?
iii) The SJF algorithm is supposed to improve performance, but notice that we chose to run process P1 at time 0 because we did not know that two shorter processes would arrive soon. Compute what the average turnaround time will be if the CPU is left idle for the first 1 unit and then SJF scheduling is used. Remember that processes P1 and P2 are waiting during this idle time, so their waiting time may increase. This algorithm could be called future-knowledge scheduling.

(c) 5 What is dispatcher? Explain the issues related to multiprocessor and multicore scheduling. 2

(d) Consider the following set of processes, with the length of the CPU burst given in milliseconds: 5

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order $P1$, $P2$, $P3$, $P4$, $P5$, all at time 0.

i) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, nonpreemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
ii) What is the turnaround time of each process for each of the scheduling algorithms in part a?
iii) What is the waiting time of each process for each of these scheduling algorithms?
iv) Which of the algorithms results in the minimum average waiting time (over all processes)?

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), Jul-Dec 2022

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

1. (a). Show the data acquisition block diagram with sensors. Explain the function of op-amp in data acquisition. 3+2
   (b) Draw the circuit diagram of an inverting amplifier with R_f= 150 KΩ; R_i= 5 KΩ; R_L= 20 KΩ and E_i = 3V. Calculate i) I; ii) V_o ; iii) A_{CL} ; iv) I_o. 6
   (c) Show the circuit symbol of IC 741 op-amp with proper pin numbering. 3

2. (a). Draw the circuit of noninverting adder and write down the working principle of noninverting adder. 5
   (b) Design a circuit to differentiate the signals where R_f= 30 KΩ; C_i= 0.1μF; R_L= 15 KΩ. if the input signal is $e_i = 0.6 \sin 2000t$ volts, find output signal, V_o. 6
   c) Show the difference between open loop voltage gain and close loop voltage gain. 3

3. (a) Discuss the low pass filter with its circuit diagram. If R_i= 12 KΩ; C_i= 0.01μF what will be the cut off frequency. Show the frequency response curve of the filter with a -20-dB/decade roll off. 6
   (b) Show the internal circuit diagram of 555 timers. Explain how it works. 6
   c) What is offset voltage? 2

4. (a). Design a R-2R ladder D/A converter and explain how it works. 5
   (b) Design a 2 bit parallel comparator A/D converter. Show the comparator outputs and digital out according to the analog input. 5
   (c) Show the classifications of bipolar logic families. How figure of merit is expressed? 4

5. a) Design a 2 input DTL NAND gate and explain how it works. Calculate average power- P h_{FE}=25; R_C= 3 KΩ; R_B= 5 KΩ; R= 4 KΩ and V_{cc}= 3.6V. 6
   b) Discuss the TTL gate with totem-pole output driver. 5
   c) Show the comparison between DTL and TTL. 3

6. a) Analyze the clipper network using a silicon diode with V_T=0.7V for given input and circuits Fig. 6(a) and Fig. 6(b). Determine the output waveform of the given network. 8

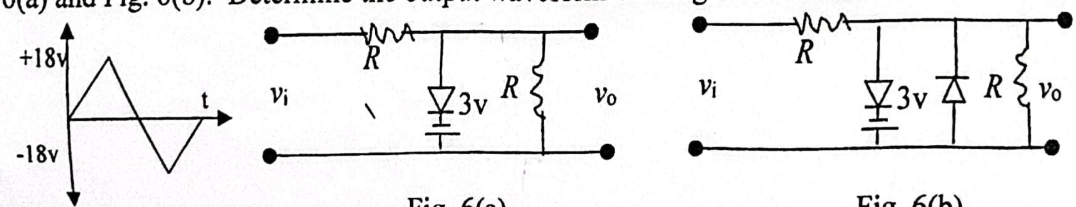
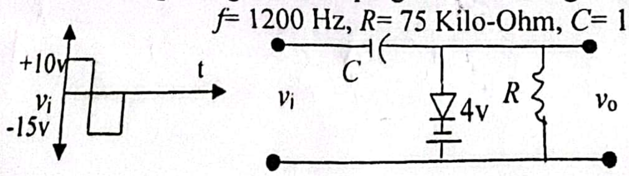


Fig. 6(a) Fig. 6(b)

   b) Analyze output voltage for given clamping network using a silicon diode with V_T=0.7V where f= 1200 Hz, R= 75 Kilo-Ohm, C= 10 μF? 6



Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2019-20

Course Code	**Course Title**	**July-December** 2022	**Credit:** 03
CCE-323	Optical Fiber Communication		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] What distinguishes an optical fiber communication system from a normal communication system? Using the relevant block diagram, describe the optical fiber communication system. 7

[B.] An 8 km length of fiber with a mean optical power launch of 120 μW will have a mean optical power of 3 μW at the fiber output. 7
Assess:

i. If no connectors or splices are present, the total signal attenuation or loss in decibels via the fiber;
ii. The fiber's signal attenuation per kilometer.
iii. the total signal attenuation for a 20 km optical link made of the same fiber with 2 k m-separated splices that each provide a 2 dB attenuation;
iv. The numerical input/output power ratio in (c).

2 [A.] What are the origins of fiber optic losses in silica? Give a thorough explanation of the intrinsic absorption loss mechanisms along with any required examples. 7

[B.] The parameters of two step index fibers are as follows: 7

i. a multimode fiber with an operational wavelength of 0.82 μm, a relative refractive index difference of 3%, and a core refractive index of 1.500;
ii. a single-mode fiber with an 8 μm core diameter, an operating wavelength of 1.55 μm, a relative refractive index difference of 0.3%, and the same core refractive index as (a).

Estimate the critical radius of curvature at which large bending losses occur in both cases.

3 [A.] i. What is optical fiber? Show its typical configuration with necessary diagram. 7
ii. Describe different advantages of optical fiber communication.

[B.] Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7\times10^{-11} m^2N^{-1}$. The refractive index and the photo elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm. Boltzmann's constant is $1.381\times10^{-21} JK^{-1}$. 7

$\lambda_R = \dfrac{8\pi^3 n^8 p^2 \beta_c K T_F}{3\lambda^4}$

4 [A.] What is meant by the fusion splicing of optical fibers? Explain the basic arc fusion method with pre-fusion method for accurately splicing optical fibers. Draw Electric arc fusion splicing diagram. 6

[B.] Briefly describe the major reasons for the cabling of optical fibers which are to be placed in a field environment. State the functions of the optical fiber cable. 3

[C.] Draw and explain a degenerate p–n junction where the position of the Fermi level and the electron occupation (shading) with applied bias and with no applied bias will be clearly depicted. 5

5 [A.] Define the two processes by which light can be emitted from an atom. Discuss the requirement for population inversion in order that stimulated emission may dominate over spontaneous emission. Illustrate your answer with an energy level diagram of a common non-semiconductor laser. 4

[B.] Explain mechanical splicing techniques that utilize grooves to secure optical fibers during the jointing process, and how these grooves contribute to the overall stability and alignment of the splice? 5

[C.] What is the main difference between LED and Laser? Draw and distinguish among different type of misalignment in fiber joint which are source of loss at a fiber-fiber connection. 5

6 [A.] Describe, with the aid of suitable diagrams, the major strategies and structures utilized in the fabrication of single-frequency injection lasers (The double-heterojunction injection laser). Indicate the reasons for the great interest in such devices. 5

[B.] A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device. 3

[C.]
- i. Draw schematic illustration of the vapor-phase deposition techniques used in the preparation of low-loss optical fibers. 6
- ii. Briefly explain outside vapor-phase oxidation process.

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2.25×10^{-2}

Dept. of Computer and Communication Engineering
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Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2019-2020

Course Code	**Course Title**	**July-December** 2022	**Credit:** 03
CCE-321	Computer Peripheral and Interfacing		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives? 4

[B.] Explain the structure of a potentiometer sensor used for measurement of linear displacement with electric circuit diagram and proper equation. List the applications of potentiometer sensor in/around your home and office/university. 5

[C.] What is Human Computer Interaction? Briefly explain most important components of HCI. 5

2 [A.] What are the different types of IoT security measures, and what are some effective strategies or best practices for securing IoT devices to protect them from potential threats and vulnerabilities? 3

[B.] How does a capacitive sensor operate to measure linear displacements, and what are the typical applications where such non-contact sensors are preferred over other types of displacement sensors? Additionally, discuss the range of displacements that capacitive sensors are capable of measuring and the factors that affect their accuracy. Explain all with appropriate diagram. 4

[C.] Imagine you're designing a real-time system for a futuristic smart city where various subsystems need to interact seamlessly to ensure optimal performance. In this context, how would you incorporate both periodic and aperiodic task models to manage different types of operations? 4

[D] Write down Norman's Seven Principles. 3

3 [A.] Define nonlinearity error. How it is different than Hysteresis error? 3

[B.] What are the key components and characteristics of a digital ecosystem, and how do these elements interact to create value within industries such as finance, healthcare, and e-commerce? Additionally, discuss the challenges and opportunities associated with managing and sustaining a digital ecosystem in a rapidly changing technological environment. 3

[C.] What is a LVDT? How does LVDT work step by step? Explain with LVDT construction diagram. 5

[D.] What are the definitions and characteristics of active and passive transducers, and how do they differ from each other? Additionally, what are the key distinctions between sensors and transducers in terms of their functions and applications? 3

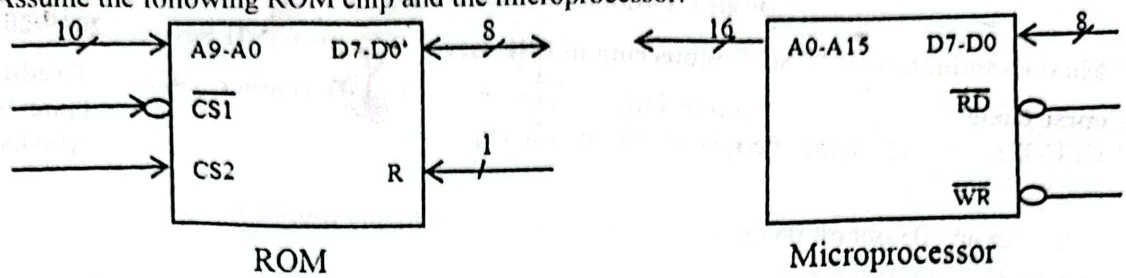
4 [A.] Show the pin configuration of 8255 PPI and separately describe the operation performed by each unit. 5

[B.] Illustrate the key components of a laser printer and provide a brief description of the function each component performs. 4

[C.] Convert -1313.3125_{10} to IEEE 32-bit floating point format. 3

[D.] Mention the main functions of an interface. 2

5 [A.] Assume the following ROM chip and the microprocessor: 5



Connect the microprocessor to the ROM to obtain the memory map: 0000_{16} through $03FF_{16}$. Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

[B.] When you press a key on your computer, you are activating a switch. There are many different ways of making these switches. Describe any four of them. 4

[C.] Explain in detail the working principle of an LCD monitor and how its components interact to produce images on the screen? 3

[D.] What is key debouncing? Why is it needed? 2

6 [A.] i. If we want to cut a 512 x 512 sub-image out from the center of an 800 x 600 image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image? 5

ii. Find the CMY coordinates of a color at (0.2, 1, 0.5) in the RGB space.

iii. If we use direct coding of RGB values with 2 bits per primary color, how many possible colors do we have for each pixel?

iv. What is the pitch of a color CRT?

v. Why do many color printers use black pigment?

[B] How can we assign input and output port of 8255? 4

[C] In USB communication, a single transaction involves the transmission of up to three packets. Mention the names of these packets and describe the specific services each one provides during the communication process. 3

[D] Provide a detailed comparison of Random Scan Displays and Raster Scan Displays, focusing on their operational mechanisms, image rendering techniques, applications, and overall performance characteristics. 2

Patuakhali Science and Technology University
Department of Computer Science and Information Technology
6th Semester (Level-3, Semester-II), **Final Examination** of **B.Sc. Engg.(CSE)**, July-December/2022,
Session: 2019-20

Course Code: CIT-323 **Course Title:** Simulation and Modeling
Full Marks: 70 **Credit hour:** 3.00 **Duration:** 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any **5** *of the following questions.*

1. a. Define system with example. Distinguish analytical solution versus simulation. Demonstrate different ways in which a system might be studied. 05

   b. Explain the next-event time-advance approach illustrated for the single-server queueing system. 04

   c. List the steps of simulation for a single-server queueing system? Consider a single-server queuing system, there are several IID random variables interarrival times, service times, and customer delays. Describe the problem statement of the single-server queuing system. 05

2. a. How to compute variance, covariance, and correlation for simulation study. Why stochastic process is used to simulate the output data. Illustrate correlation function of the process of delays in queue D1, D2, ...for the M/M/1 queue. 05

   b. Define verification, validation, and credibility in simulation model. Mention the issues in an accreditation decision. Show the timing relationship of validation, verification, and establishing credibility. Derive the mathematical formulation in a good estimate of the mean of the system using both validation and output analysis. 06

   c. Explain the inspection approach of statistical procedure for comparing real-world observation and simulation output data. 03

3. a. List some sources of randomness for common simulation applications. Write the properties of location, scale, and shape parameters of continuous distributions 04

   b. Many random number generators in use today are linear congruential generator, LCGs. Explain, how to obtain the desired random numbers from LCGs. What is inverse transform function? Let X have the exponential distribution with mean β, Generate the desired random variate using the inverse transform function. 06

   c. What is the necessity of statistical analysis on simulation output data? Describe and illustrate transient and steady-state behavior of a stochastic process. 04

4. a. Define computer simulation models. Describe the steps in simulation study. 02

   b. In a pure pursuit, there is a target bombers which moves along a predetermine path (given on table below) and there is a pursuer aircraft who follows the target bombers, redirecting itself towards the target bombers at fixed intervals of time.

Target bombers moving path:

Time,t	0	1	2	3	4	5	6	7	8
xb(t)	160	161	167	172	181	189	199	215	239
yb(t)	0	4	9	11	14	24	34	41	52
Time,t	9	10	11	12	13	14	15		
xb(t)	245	254	265	271	289	293	300		
yb(t)	45	41	36	39	34	28	12		

12

The fighter aircraft following the enemy bombers with speed at 30KM/min to destroy it. The fighter aircraft is at position *xf, yf* (50, 50) when it sights the bomber that is at time t=0, the time at which the pursuit begins. The fighter corrects its direction after a fixed interval of one minute, so as to point towards a bomber and shoots the bomber by firing a missile as soon as it is within a distance of 10 KM. The pursuit ends. In case the bomber is not shot within 15 minutes of the pursuit, the pursuit is abandoned. Simulate the problem to determine the pursuit is end/abandoned.

5. a. Define the simulation of continuous system. Describe the disadvantages of analog simulation. 02

b. Why the random numbers generated by computer are called pseudo random numbers? Discuss the pitfalls of congruence method of generating random numbers. 02

c. Consider a chemical reaction of two gas H_2 and O_2 react to produce H_2O. The rate of the reaction is depending upon the ratio in which H_2 and O_2 are mixed and some other factor like temperature, pressure and humidity which remains constant during the reaction. There will also happen backward reaction where H_2O decomposes back into H_2 and O_2. If C_1, C_2 and C_3 are the amount of H_2, O_2 and H_2O at any instant of time t, the rate of increase of H_2, O_2 and H_2O are given by the following differential equations: 10

$$\frac{dC_1}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_2}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_3}{dt} = 2K_1C_1C_2 - 2K_2C_3$$

Where, K_1 and K_2 are constants. If K_1=0.025, $K_2 = 0.01$ and at time $t=0, C_1 = 50.00\ m^3$, $C_2 = 25.00\ m^3$ $C_3 = 0.00\ m^3$, $\Delta t = 0.2\ minute$. Simulate the experiment for 3 minute to determine the amount of H_2, O_2 and H_2O.

a. Define random variables. What are the purposes of testing the uniformity and independence of random values? 02

b. Define autocorrelation. A sequence of 100 random numbers is given below. Use Chi-Square test with 99% confidence level. Test the autocorrelation of the given numbers employing the chi-squared test. 07

21	81	92	23	96	20	68	57	79	84
82	62	12	08	92	83	74	85	60	49
48	37	65	74	22	11	28	10	55	82
72	95	08	85	79	95	86	11	16	52
70	55	50	87	67	51	72	38	29	62
71	12	07	75	56	34	40	67	24	86
18	82	41	29	63	06	84	01	20	06
06	33	14	79	25	65	57	47	74	68
54	35	81	07	88	96	70	85	29	13
12	91	26	57	30	22	90	03	13	31

N.B: You can use chi-squared table below as required.

c. Perform the Kolmogorov-Smirnov test to testify the uniformity of following random numbers at 95% level of significance. 0.37, 0.1, 0.5, 0.88, 0.37, 0.29, 0.84, 0.78, 0.08, 0.46 at $\alpha = 0.01$ and N= 10, critical values is 0.368; At $\alpha = 0.05$ and N= 10, critical values is 0.410. 05

Table: Chi-square Distribution

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57

15 7 63 6 11 9 13 9 15

17 6 17 6 12 9 13 9 10

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Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology(CSIT)
Final Examination: July-December 2022
Course Code: CIT 322 | Course Title: Operating Systems Sessional
Session: 2019-20, Program: B.Sc. Engg.(CSE), Semester: 6th

Marks – 70

[Answer the marked questions from Set A , Set B]

SET-A

1. The following processes are being scheduled using a FCFS scheduling algorithm.[USE CONCURRENT PROCESS] 25

Process	Priority	Burst Time	Arrival Time
P1	2	15	0
P2	1	20	0
P3	4	20	20
P4	2	20	25
P5	3	5	45
P6	2	15	55

   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?
   iii.

2. The following processes are being scheduled using a preemptive, SJF scheduling algorithm.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   Show the execution of these processes using the following scheduling algorithms: **SJF**.
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?
   iii.

3. Create some process and apply round-robin CPU scheduling algorithm on them.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?

4. Create some process and apply priority CPU scheduling algorithm on them.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?

5. Create some process and apply First Come First Serve (FCFS) CPU scheduling algorithm on them.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), Jul-Dec 2021

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

1. (a). Explain how transistor is used as inverter? Draw the circuit of a CMOS switch. 3+2

(b). Discuss on speed of operation, noise immunity and figure of merit in digital logic families. 3

(c). Draw the circuit diagram of 3 inputs RTL NOR gate and explain how it works. 6

2. (a). For the DTL NAND gate where diode and transistors parameters are given as
Diode: Voltage across the diode=0.7V; Cut-in voltage=0.6V
Transistor: Cut-in voltage=0.5V; $V_{BE,sat}$=0.8V ; $V_{CE,sat}$=0.2V ; h_{FE}=20; R_C= 2 KΩ; R_B= 4 KΩ
Resistor connected with V_{cc}=4 KΩ. Calculate i) fan-out ii)noise-margins iii) average power- P. 6

(b). How 3-input TTL NAND gate is operated in digital logic families? 4

(c) Discuss about the saturated bipolar logic families. 4

3. (a) Show the pin configuration of a typical op-amp with its function. 4

(b) Draw the circuit of an Op-Amp as voltage follower and find an expression for its voltage gain. 6

(c) Draw the circuit for a non-inverting amplifier and explain its operations. 4

4. (a). Show the differences between open loop gain and close loop gain. 3

(b) Draw the circuit diagram of integrator and differentiator and explain how it works. 6

(c) Draw the circuit diagram of subtractor and explain how it works. 5

5. (a) Find the analog output voltage of a 3-bit D/A converter for all possible inputs. Assume K=1 5

(b) Design a weighted-resistor D/A converter and explain how it works. 5

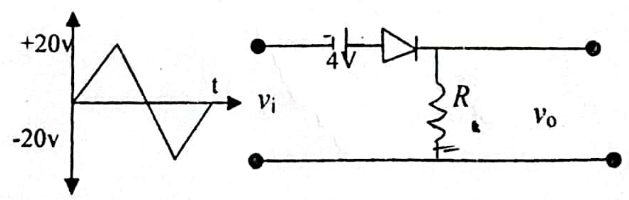
(c) How offset voltage is used in D/A converter? 2

(d) What is quantization error? 2

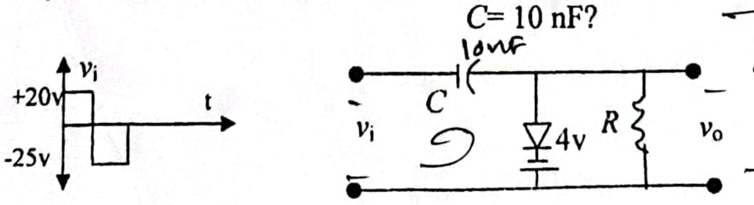
$v_i = \frac{R_1 R_2 \times v_o}{R_1 \times R_2}$

$v_o =$

6. (a) Analyze the clipper network with given input and determine the output waveform of the given network. 6



(b) Analyze output voltage for given clamping network where f= 2500 Hz, R= 25 Kilo-Ohm, C= 10 nF? 6



(c) How resistors and capacitors are considered for clampers? 2

$v_i = v_o$

Inotry

Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Final Examination, July-December-2021

Program: B. Sc. Engg.(CSE)
Session : 2018-19

Course Code : CIT-323 | Course Title : Simulation and Modeling

Full Time: 3 Hours | Full Marks: 70

[Figures in the right margin indicate full marks]
Answer any 5 of the following questions.

1. a. Define the following terminologies: i) System ii) Model iii) Simulation iv) Analytical solution. Demonstrate different ways in which a system might be studied. 04
   b. What are the steps of simulation for a single-server queueing system? Describe the problem statement and intuitive explanation of a single-server queueing system. 10

DDP MM VIP S

2. a. What is random variable? Compute variance and covariance based on a set of random variates. Why stochastic process is used to simulate the output data? Illustrate correlation function of the process of delays in queue D1, D2, ...for the M/M/1 queue. 07
   b. Explain hypothesis testing including null and alternative hypothesis for mean. What is the difference type I error and type II error? Explain and illustrate the strong law of large number. 07

3. a. Write the properties of location, scale, and shape parameters of continuous distributions. What is inverse transform function? Let X have the exponential distribution with mean β, Generate the desired random variate using the inverse transform function. 07
   b. What is Monte Carlo Method? Write down the difference between monte-carlo and stochastic simulation. 02
   c. Define Monte-Carlo method. Solve the integrals $I = \int_1^5 \frac{x^4}{3} dx$ by monte-carlo method. [USE Random Number 48, 36, 44. 41, 46, 18, 32, 41, 23, 23 for X coordinate and Random Number .52, .18, .30, .88, .21, .25, .57, .82, .9, .63 for Y coordinate, you have a right to do scaling, if required] 05

4. a. Define computer simulation models. Write down the qualities of an efficient random number generator. 02
   b. A machine center has three identical bearings, which fail according to the following probability distribution in table A. The present maintenance policy is to change a bearing as and when it fails. When bearing fails machine center stops, a technician is called to replace the failed bearing with a new one. The time between the failure of the bearing and reporting of the technician (delay time) is random and is distributed as table B. 10

Table: A

Bearing Life(Hours)	*Probability*
1200	0.10
1400	0.25
1600	0.30
1800	0.25
2000	0.10

Table: B

Delay Time(Minutes)	*Probability*
2	0.3
5	0.5
8	0.2

It takes 15 minutes to change one bearing, 25 minutes to change two bearing and 35 minutes to change all three bearings. Downtime of the system costs $5 per minutes, direct on job costs of the technician is $30 per hour and the cost of bearing is $30. The maintenance department is interested in evaluating an alternative policy of replacing all the three bearings, whenever a bearing fails. Simulate the present and proposed policy for 10000 hours and choose the better policy.

   c. What is normally distributed random numbers? Describe the properties and applications of normally distributed random numbers. 02

5. a. Define the simulation of continuous system. Describe the disadvantages of analog simulation. 02

b. There are a number of moving objects, which chase each other in a serial order like A chases B, B chases C, and C chases D etc. Each object moves towards its target unmindful of the fact that it itself is being targeted by some other object. Depending upon the original location, and speed of motion of the object, each object will take its own time to hit its target. As soon as hit occurs, the chase ends. The initial location of the objects A, B, C, and D are (0,0), (0,10), (0,20), (0,30) respectively and their velocities are 30km/hr, 25km/hr, 20 km/hr and 15km/hr respectively. Object D moves towards a fixed at (30,50), while C moves always in a direction pointing towards D, the object B moves always pointing towards C and object A always pointing towards B. It can be assumed that when the distance between two objects less than 0.005 hit is taken to occur. Identify, which object will hit the target at first and in what time? 10

c. Why the random numbers generated by computer are called pseudo random numbers? Discuss the pitfalls of congruence method of generating random numbers. 02

6. a. Define random variables. What are the purposes of testing the uniformity and independence of random values? 02

b. i. A sequence of 100 random numbers is given below. Use Chi-Square test with $\alpha = 0.05$ to test these numbers are uniformly distributed. 12

21	81	92	23	96	20	68	57	79	84
82	62	12	08	92	83	74	85	60	49
48	37	65	74	22	11	28	10	55	82
72	95	08	85	79	95	86	11	16	52
70	55	50	87	67	51	72	38	29	62
71	12	07	75	56	34	40	67	24	86
18	82	41	29	63	06	84	01	20	06
06	33	14	79	25	65	57	47	74	68
54	35	81	07	88	96	70	85	29	13
12	91	26	57	30	22	90	63	13	31

ii. Define autocorrelation. Test the autocorrelation of the given numbers (6(b)i) by employing the chi-squared test with 99% confidence level.

N.B: You can use chi-squared table below as required.

Table: Chi-square Distribution

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57

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Dept. of Computer and Communication Engineering
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Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2018-2019

Course Code	**Course Title**	**July-December** 2021	**Credit:** 03
CCE-321	Computer Peripherals and Interfacing		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives? 4

[B.] What is communication model in IoT? Explain different types of communication model in IoT with diagram and example. 6

[C.] Write down the difference between Embedded and Real Time Systems. 4

2 [A.] What is the periodic and aperiodic task model in real-time system? Explain with example. 4

[B.] What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram. 4

[C.] How many principles are there in Norman design principles? Write down Norman's design principles for IoT. 3

[D] Briefly explain the most 6 communication protocols used by IoT. 3

3 [A.] Explain the characteristics of RTS. 3

[B.] Write down the difference between Sensors and Transducer. 3

[C.] What is Human Computer Interaction? Briefly explain most important components of HCI. 5

[D.] What are the types of IoT security? How to Secure IoT Devices? 3

4 [A.]
a. What do you mean by true color representation? Does it represent any disadvantages? How to handle the problems? 5
b. If we use 5 bits each for red and blue and 6 bits for green in direct coding method, how many possible simultaneous colors do we have?
c. Interpret EFE_{16} as unsigned, signed, and floating point numbers that a microprocessor can understand. 4
d. What are the common problems with scanner? Share some tips to troubleshoot the basic scanner related problems. 3
e. Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port. 2

5 [A.] How do the USB peripherals communicate with microprocessor? Give the pin-out for USB type C 3.0. 5

[B.] Categorize and explain the operational modes of 8279. How many ways the keyboard is interfaced with the CPU? 4

[C.] The microcomputer communicates with an external device via one or more registers called I/O ports using programmed I/O. Illustrate the functions of two of them with brief description. 3

[D.] Make a comparison between buffers and spoolers. 2

6 [A.] Suppose, a processor is having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor. 5
a. What is the memory capacity of the processor?
b. How many address lines are required for a RAM?
c. How many RAM chips are required?
d. What will be the size of address decoder?

cn 16×10
∴ no $2^{12} \times 16$

[B.] Show the software keyboard interfacing process through a flowchart. Describe each step for a 4×4 keyboard. 4

[C.] Serialize the steps associated in turning on or off a pixel in LCD monitor. 3

[D.] An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer. 2

$2^{112} - 2^{4}$

for 1 11011 11110

$= 2^{8}$

256